

Course Syllabus

CSC 5666: *Advanced Machine Learning*, Fall 2026
Professor Austin P. Wright

I Course Information and Contacts

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Office Hours (In person at 14-222):

- Mondays 11-12, and Mondays, Tuesdays, and Wednesdays from 1pm-2pm.
- By appointment at <https://calendly.com/awrigh20-calpoly/30min>.

Course Homepage:
Gradescope: <https://www.gradescope.com/courses/1355369> | **Entry Code:** YB3GJJ
Edstem: <https://edstem.org/us/join/nMfdkZ>

Contents

I	Course Information and Contacts	1
II	Learning Objectives	1
III	Course Overview	1
IV	Assessment and Grading	3
	Participation	3
	Final Project	3
V	Schedule	4
VI	Policies	5
	Classroom Conduct	5
	Attendance	5
	Course Notes	5
	GenAI/LLM Use	6

II Learning Objectives

As an advanced graduate course the ultimate learning object of this course is for you to develop the knowledge and skills to engage with and contribute to contemporary research in machine learning. Of course, as this field is rapid changing and very large we will cover some of the most important and fundamental approaches of current relevance and interest, while simultaneously developing the research skills to bring yourself up to speed in any other particular sub-area in the future.

III Course Overview

This course will be based structurally around the textbook: *Probabilistic Machine Learning: Advanced Topics*, Kevin Murphy, 2023¹, alongside additional research papers depending on topic.

Most machine learning research is about mapping inputs to outputs: approximating functions $f : \mathcal{X} \rightarrow \mathcal{Y}$, typically with deep learning — and prediction of this kind is sometimes dismissed as little more than sophisticated curve fitting. That charge is not entirely fair: once $\mathcal{X}$ and $\mathcal{Y}$ are themselves high-dimensional (images, sentences, graphs, sequences of actions), the intuitions that make “curve fitting” sound simple in one dimension stop applying. But the criticism still points at something real: current approaches are weighted heavily toward predicting observable patterns, and comparatively light on **understanding** the latent structure that produces those patterns.

That deeper understanding is not just an intellectual nicety. It is necessary for science, where uncovering root causes and mechanisms — e.g. behind a disease —

¹The book can be found for free at: <https://probml.github.io/pml-book/book2.html>

is the actual goal, not merely predicting an outcome. It is also necessary for building systems that are **robust**, meaning they keep working under unexpected shifts in the data distribution (a central concern in fields like robotics), and **efficient**, chiefly in the sense of learning well from small amounts of data. Data efficiency matters because data is not equally abundant everywhere: domains like healthcare and robotics are comparatively data-poor, even as language and vision have become data-rich through internet-scale scraping. Computational efficiency matters too, but is treated as secondary given the continued growth of available compute.

The approach in these notes, following the textbook, is **model-based**: rather than fitting $f : \mathcal{X} \rightarrow \mathcal{Y}$ directly, we try to learn parsimonious representations of the underlying **data-generating process** (DGP) itself. This mirrors the scientific method, where observations are explained by developing theories, and it is formalized as **Bayesian inference** applied to probabilistic models — treating a model’s parameters and structure as unknown quantities, and updating beliefs about them as data arrives.

Once a toolbox of inference methods is available — sometimes as simple as a maximum-likelihood point estimate obtained by optimization, sometimes considerably more involved — attention turns to the choice of model itself, which depends on the task, the kind and amount of data at hand, and the metric of success. Four broad task types organize the material that follows:

- **Prediction** — conditional models $p(y|x)$, $x \in \mathcal{X}$ (often high-dim), $y \in \mathcal{Y}$ (often low-dim); one right answer, possibly uncertain.
- **Generation** — models $p(x)$ or $p(x|c)$ for optional conditioning c ; multiple valid outputs, harder to evaluate.
- **Discovery** — latent-variable models $p(z, x) = p(z)p(x|z)$; compute $p(z|x)$ to uncover hidden structure.
- **Control** — decision-making under uncertainty, leading into causality.

This course follows the same arc, adapted to a single-quarter schedule. Having covered probability and statistics here, the next weeks complete the remaining **fundamentals**: probabilistic graphical models (directed and undirected) for representing structured joint distributions compactly, and information theory (entropy, KL divergence, mutual information) alongside a first pass at Bayesian optimization. With that vocabulary established, we assemble a toolbox of **inference** algorithms — message passing, variational inference, and Monte Carlo methods including Markov chain Monte Carlo — the general-purpose machinery used to compute or approximate posteriors throughout everything that follows.

We then turn to models for **prediction**: conditional distributions of the form $p(y|x)$ that assume a single, possibly uncertain, right answer. This course develops prediction at its most Bayesian, through Bayesian neural networks, which place a distribution over network weights to quantify predictive uncertainty, and Gaussian processes, which place a distribution directly over functions.

Next come models for **generation**: distributions of the form $p(x)$ or $p(x|c)$, for which many outputs can be valid and evaluation is correspondingly harder than in

the prediction setting. This spans variational autoencoders, autoregressive models, energy-based models, diffusion models, and generative adversarial networks.

We then turn to **discovery**: latent-variable models of the form $p(z, x) = p(z)p(x|z)$, whose goal is to compute $p(z|x)$ and uncover meaningful hidden structure — covered here via latent factor models and latent state-space models, together with Bayesian nonparametric methods that let model complexity grow with the data. Representation learning and interpretability pursue the same goal from the other direction, asking what structure a predictive model has learned implicitly, without positing an explicit generative model of the data.

Finally, the course closes where the textbook does: with causal inference — structural causal models, interventions, and counterfactuals — which asks not just what a model predicts, but what would happen under a hypothetical change to the system. As in the book, the aim throughout is not encyclopedic coverage but working command of the fundamentals, deep enough in places to touch the research frontier, sufficient to keep learning independently in whatever direction the field moves next — the final project is where that independence gets exercised.

IV Assessment and Grading

Your grade for this course will be based on two kinds of assessment, each accounting for half of your final grade, participation and your final project.

Participation

In this course there will be many opportunities for interaction, participation, and in class exercises. Your participation in these activities is what makes the class work, not just for you, but also for everyone else in the class. Participation is expected and so a portion of your final grade is dependent on your engagement. Your participation grade does not depend at all on whether you get questions correct or how good your in class activity submissions are, it only depends on how regularly you are able to show up and engage with the material and with the rest of the class. While perfect attendance is not required to score a 3 on this assessment², since nobody is immune to the slings and arrows of outrageous fortune, I still expect your best effort to make the class work.

Final Project

The most interesting single thing you will do in this course will be your final project. In this project you will have substantial freedom to choose what you work on. You will complete the final project in teams of 2-4 students, and so I highly recommend you get to know the other students in the class, find anyone with similar interests, and get started as early as possible. Throughout the course there will be many check-in meetings to help get feedback and keep on pace, as well as intermediate deliverables.

²Nor would it be a guarantee as just showing up is not the same thing as participating!

V Schedule

Below is a tentative (and probably too optimistic) schedule of the topics we will cover in the course. As the semester progresses things may change, so pay attention to announcements/notifications/emails.

Week	Date	Lecture Topic	Deadlines
1	Mon, Aug 24	Course Introduction	
1	Wed, Aug 26	Probability Foundations	
2	Mon, Aug 31	Probabilistic Graphical Models	
2	Wed, Sep 2	Probabilistic Graphical Models Continued	
3	Wed, Sep 9	Information Theory Foundations	
4	Mon, Sep 14	Inference	
4	Wed, Sep 16	Variational Inference	
5	Mon, Sep 21	Monte-Carlo Methods Overview	
5	Wed, Sep 23	Markov Chain Monte-Carlo	
6	Mon, Sep 28	Bayesian Neural Networks	Project Proposal
6	Wed, Sep 30	Gaussian Processes	
7	Mon, Oct 5	Generative Models Overview	
7	Wed, Oct 7	Variational Auto-encoders	
8	Mon, Oct 12	Auto-regressive Models	
8	Wed, Oct 14	Energy Based Models	Literature Review
9	Mon, Oct 19	Diffusion Models	
9	Wed, Oct 21	Generative Adversarial Networks	
10	Mon, Oct 26	Latent Factor Models	
10	Wed, Oct 28	Latent State Space Models	
11	Mon, Nov 2	Bayesian Nonparametrics	
11	Wed, Nov 4	Representation Learning	Prototype Check-in
12	Mon, Nov 9	Interpretability	
12	Fri, Nov 13	Interpretability Continued	
13	Mon, Nov 16	Causal Inference	
13	Wed, Nov 18	Causal Inference Continued	
Fall Break			
14	Wed, Dec 2	Special Topics	
15	Mon, Dec 7	Final Paper Review	Final Paper Draft
15	Wed, Dec 9	Final Project Presentations	Final Presentation
Finals	Fri, Dec 18, 1:00–3:30 PM	Retakes	Final Paper

VI Policies

Classroom Conduct

Our classroom and lab are to be places of learning and inclusion. Students of all ages, abilities, background, race, sexual orientations, beliefs, religious affiliations, gender identities, and origins are to be treated with dignity and respect as contributors to our scholarly environment. Recognizing the following points is a non-negotiable prerequisite to participate in this course:

- ***We recognize that every single student in the class belongs here.*** We all have different backgrounds and experiences and we are not snobs about which backgrounds do or don't count.
- After our work is complete, we prioritize the education of others and actively offer to help, explain, debug, etc. in order to support one another's learning. We do not share our working solution, but explain the logic/thinking behind our solution and help others recognize errors in their implementation when invited to do so.
- We consistently make the effort to recognize and validate multiple types of contributions to a positive classroom environment.

Attendance

My goal is for lectures to be interactive which only works when people show up. There will be frequent small group discussions, and I am also likely to call on individual students. If I call on you, it's totally okay to get an answer wrong or to not know the answer³. However, if being called on is likely to be uncomfortable or disruptive for you, let me know.

I don't allow the use of laptops during lecture sessions without special dispensation. There is plenty of evidence that suggests that laptops and other devices are distracting not only to the student using them, but also to those around them. Additionally, taking handwritten notes tends to lead to better learning outcomes⁴. If you need a laptop to take notes in class, please talk to me. You will have the opportunity to use your computer to work during specifically set aside time.

³ Indeed, this is probably a sign that I have moved too quickly or been unclear about something. But I can only learn that and make adjustments if people are in class and able to speak up.

⁴ <https://journals.sagepub.com/doi/abs/10.1177/0956797614524581>

Course Notes

My goal is to make my course/lecture notes and/or slides available to you as a study resource. They will be able to be uploaded after class and available on the course website. However, it is essential that you realize that these are not complete or sufficient to replace taking your own notes, but rather they may help you structure your notes and studying.

The primary purpose of the course notes and slides is to help *me* structure and keep on track in lecture, but of course *I already know the material* and so what I need to have written in order to lecture is very different than what you need to write to internalize the material for the first time. You are still responsible for the content actually covered in lecture.

GenAI/LLM Use

The goals for this course are for *you* to master the material, which can only be done if *you* are the one doing the work. Therefore I must emphasize that for this class ***you cannot use Generative AI or Large Language Models for any aspects of writing or design.*** This class is meant to develop your research and analysis skills, and as such I want to see *your* research and analysis. If it is clear through the oral components of the class that the work you are submitting is not your own, this will have consequences on your grade and may result in academic integrity violations.

That being said, this course does not have a learning objective of teaching you how to code⁵. Therefore use of AI development tools for writing code and implementation is allowed. However, be very careful, as you will still be completely responsible for everything you submit and it is a very slippery slope. We learn best by struggling and surmounting challenges. Uncritical reliance on GenAI tools will short-circuit this process. Sure, you will get an answer quickly, but the answer is not our objective; our objective is the process that gets us there. (Just like the goal of lifting weights in the gym is not just to have the weights in the air.)

⁵This is assumed as a prerequisite for the course

If you do use AI assistants to help you study, you're encouraged to put them in "study mode" first. Different companies have different names for this:

- "Study mode" in ChatGPT
- "Learning mode" in Claude
- "Guided learning" in Google Gemini

These "modes" nominally do not jump straight to an answer, but try to lead you to an answer while helping you build your understanding. Even still, be very wary of these tools even in a guard-railed state. Think very hard about how, if you do not develop the fundamental skills and you need such tools in order to succeed, what your ultimate value is after graduation.